

Natural and modified (1-->3)-beta-D-glucans in health promotion and disease alleviation.

A number of polysaccharides with beta-glycosidic linkage are widespread in nature in a variety of sources. All have a common structure and the (1-->3)-beta-D-glucan backbone is essential. They have attracted attention over the years because of their bioactive and medicinal properties. In many cases their functional role is a mystery, in others it is well established. Because of their insoluble chemical nature, particulate (1-->3)-beta-D-glucans are not suitable for many medical applications. Various methods of changing or modifying the beta-D-glucan chemical structure and transforming it to a soluble form have been published. The beta-D-glucan bioactive properties can be affected positively or negatively by such modifications. This review examines beta-glucan sources in nature, health effects and structure-activity relationships. It presents the current state of beta-D-glucan solubilization methods and discusses their effectiveness and application possibilities for the future.